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EVALUATION OF PREDICTION METHOD OF TARGET WORDS USING INDUCTIVE LEARNING

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ABSTRACT

This paper presents and evaluates the prediction method of target words for unknown words with units of character strings. The system combines the units and generates the target words. We made the evaluation experiment for the data which are derivative words and compound words from four categories of English corpus. The system translates English into Japanese. In this experiment, the average rate of effectiveness is about 63.3%. In order to evaluate the usefulness of inductive learning, we made another experiment on the system without inductive learning and the results of the first experiment have been compared with the results of another experiment. The maximum difference of the rates of effectiveness between two experiments is about 25.9 points and we confirm the usefulness of inductive learning in our method.

keywords: machine translation, inductive learning, prediction, target word, unknown word.

1 INTRODUCTION

Machine translation systems could be the effective tool and various approaches have been proposed [1]. On the other hand, we have not practicably employed the machine translation system because it has a lot of problems to resolve. One of them is that it cannot translate the unknown words which are not registered in the dictionary of the system.

Knight et al.[2] proposed and evaluated the method of transliteration from Katakana phrases in Japanese into target words in English. Knight et al.[2] translate proper names and technical terms across languages with different alphabets and sound inventories. However the unknown words are not limited to proper names and technical terms. Kumano et al.[3] proposed the method to generate the dictionary from parallel texts and Kitamura et al.[4] presented the automatic extraction method of a translation dictionary from parallel corpora. Kumano et al.[3] and Kitamura et al.[4] use the parallel text and require the good correspondence between source and target language. However it is difficult to prepare such parallel text on many different fields.

Major approaches on natural language processing have been called Rule-based approach. Rule-based systems use the rules which were given at the beginning. However they have some problems. For example, machine translation systems cannot translate the sentences which can not be translated using given rules. Therefore the researches[5, 6] have aimed to realize a system with the capability of acquiring language and knowledge, which is similar to one in human being. Our researches try more practical applications of inductive learning and confirm the effectiveness.

In our previous research [7], we have studied the prediction method of target words and have shown the effectiveness of our method for the data which were extracted from one category "Press Reportage" in Susanne Corpus [8]. In this paper, we describe the results of evaluation experiments for the data which have been extracted from four categories of the corpus. Moreover, in order to indicate the usefulness of inductive learning, we have compared the results of the experiments on the system based on our method with the results of another experiments on the system without inductive learning. We confirm the effectiveness of our method and the usefulness of inductive learning in our method.

2 UNITS FOR THE PREDICTION

This section explains the units for the prediction process. Some of the units for prediction process are the pairs of bases and their target words and the pairs of affixes and their target words. Other units are automatically extracted from the pairs of the character strings in a source and target language using inductive learning. We have called the extracted character strings "a Piece of Word(PW)" and have called the extracted pairs of PW's " a Pair of Pieces of Word(PPW)" [7]. Figure 1 presents an example of extraction of PPW's. In the figure, italic characters express Japanese phonograms.

We consider that extracted PPW's are equal to the pairs of bases and their target words and the pairs of affixes and their target words. The reason is that the units of character strings are included in words and target words. The number of the units which we can prepare is finite. Therefore all the required units cannot be given and we need to use the automatical extraction method for the units using inductive learning.

3 OUTLINE OF OUR METHOD

3.1 OVERVIEW

Figure 2 shows the outline of our experimental system. After the input of an unknown word, the prediction process is executed. In the prediction process, at first, the system searches for the PPW's which have the English PW's which construct the character strings

Source Pair 1:

Word 1: earthen
Word 1 for translation: 土で作られた (*tsuchi de tsukura reta*)

Source Pair 2:

Word 2: oaken
Word 2 for translation: オーク材で作られた (*ohkuzai de tsukura reta*)

⇓

Pair of strings of common: en で作られた (*de tsukura reta*)
Pair 1 of strings of difference: earth 土 (*tsuchi*)
(Extracted from Pair 1)
Pair 2 of strings of difference: oak オーク材 (*ohkuzai*)
(Extracted from Pair 2)

⇓

A variable is given to the character string of common.

PPW 1: @1 en @1 で作られた (*de tsukura reta*)
PPW 2: earth 土 (*tsuchi*)
PPW 3: oak オーク材 (*ohkuzai*)

Figure 1: Examples of PPW's.

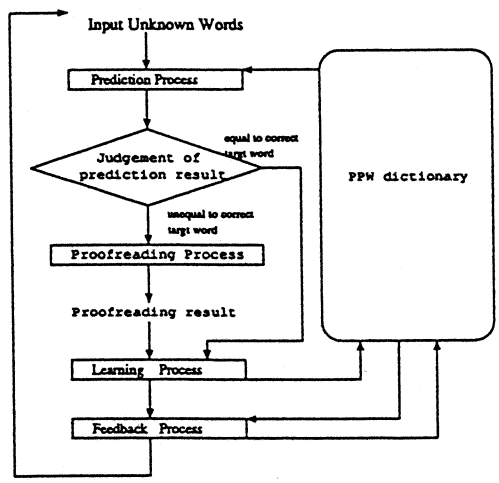


Figure 2: Outline of our experimental system.

of the unknown word. Thereafter, the Japanese PW's of the searched PPW's generate the prediction results. When two Japanese PW's are combined, one PW is substituted for the variable in another PW. In the learning process, new PPW's are acquired from the pairs of character strings using inductive learning. The PW's are extracted from common character string and the different character strings between two source character strings. The extracted pairs become new PPW's and are add to the dictionary. After the judgment whether the prediction result is correct or erroneous, in the feedback process, the numerical values with PPW's are calculated. To improves the capability of prediction, this process has been done.

3.2 PREDICTION PROCESS

In the prediction process, the system searches for PPW's whose English PW's compose the character strings of unknown word and generates the prediction results with the Japanese PW's of searched PPW's. In the case that the system could generate some available combinations of Japanese PW's, some prediction results could be generated. In the case, the system should decide the order of priority among the results and indicate the most reliable result. When the system ranks the results in order of reliableness, it refers to some numeric values of PPW's and Japanese PW's. The numeric values are as follows:

1. the number of used PPW's in the prediction result.
2. the average of appearance frequency of PPW's used in the prediction result.
3. the average of correct prediction frequency of PPW's used in the prediction result.
4. the average of erroneous prediction frequency of PPW's used in the prediction result.
5. the average of correct connection frequency of Japanese PW used in the prediction result.
6. the average of erroneous connection frequency of Japanese PW used in the prediction result.

The appearance frequency of PPW expresses how many times the PPW has been used in the prediction process. The correct prediction frequency of PPW indicates how many times the PPW has been used in the correct prediction results and the erroneous prediction frequency of PPW indicates how many times the PPW has been used in the erroneous prediction results. Furthermore the correct connection frequency of Japanese PW shows how many times the connection has been used in the correct prediction results and the erroneous connection frequency of Japanese PW shows how many time the connection has been used in the erroneous prediction results.

3.3 LEARNING PROCESS

The given PPW's are insufficient to predict words for translation of unknown words. In the learning process new PPW's are extracted from pairs of English words and Japanese words for translation and from PPW's which are registered in the PPW dictionary.

When the new PPW's are acquired, the system uses the information of the appearance of written words and words for translation. Therefore the system extracts some PPW's which have erroneous correspondences of meaning between English PW's and Japanese PW's. To reduce the erroneous correspondences, we define some conditions for the learning process. The conditions are as follows:

Condition 1 when English PW's are extracted, the common character strings have more than two characters.

Condition2 when PPW's are extracted,

Condition 2-a one source pair of English and Japanese strings is composed by one pair of common character strings and another is composed by one pair of common character strings and one source pairs of different character strings.

Condition 2-b both of pairs are composed by one common string and one different string.

4 EVALUATION EXPERIMENTS

4.1 PROCEDURE

In the experiment for evaluation, we have recognized unknown words as the words which are not regarded in the dictionary [9]. The kinds of unknown words are derivative and compound words. We have selected the data for the the experiment from four categories in Susanne corpus [8]. These categories are "Press Reportage," "Bells

Table 1: Results of evaluation experiment

Name of category	(A)	(B)	(C)	(D)	(E)	(F)
Press Reportage	31	77	60.0	72	40.0	180
Belles Letters, Biography, Memories	43	146	63.9	107	36.1	296
Learned Writing	57	178	62.0	144	38.0	379
Adventure and Western Fiction	21	117	67.6	66	32.4	204

Table 2: Results of evaluation experiment on the system without inductive learning

Name of category	(A)	(B)	(C)	(D)	(E)	(F)
Press Reportage	26	60	47.8	94	52.2	180
Belles Letters, Biography, Memories	18	117	45.6	161	54.4	296
Learned Writing	25	112	36.1	242	63.9	379
Adventure and Western Fiction	14	83	47.5	107	52.5	204

Unknown English word: rightfully
PPW 1: @1 ly @1 に (*ni*)
PPW 2: rightful 正当な (*Seitou na*)
⇓
Result : rightfully 正当な に (*Seitou na ni*)
Correct word for translation: rightfully 正当に (*Seitou ni*)

Figure 3: An Example of result of Group 2

letters, Biography, Memories," "Learned Writing" and "Adventure and Western Fiction." The numbers of data in the four categories are 180, 296, 379 and 204.

The pairs of the bases and their target words are given to the PPW dictionary of the system. We extracted the pairs from the pairs of the words and target words which are registered in the dictionary [9]. The dictionary [9] is an electric dictionary and has 49,336 English words and it may be useful for people to read general texts. We extracted 102,156 pairs of English words and their Japanese target words from the dictionary. Furthermore, the pairs of affixes and target words are also given. We consulted the book [10] to extract the PPW's. As a result, 351 PPW's are extracted. They include 69 PPW's which are the pairs of prefixes of English word and Japanese target words. And they also include 282 PPW's which are the pairs of suffixes of English word and Japanese target words. In this experiment, the total number of initial PPW's is 102,507.

We have classified the results of prediction into three groups. We define the correct target word of the unknown word as the word registered in the English-Japanese dictionary [11]. The classification of results is described below.

Group a The result is equal to the character strings of the correct target word, or the only one part in the result is rearranged by the equivalent string.

Group b The result is equal to a part of the base of the correct target word.

Group c The result cannot be classified into neither Group a nor b.

Figure 3 shows an example of prediction result of Group b. The prediction result "正当な に (*Seitou na ni*)" is equal to a part of the appearance of written correct words for translation"正当な (*Seitou na*)".

We define the rate of effectiveness. The value expresses the effectiveness in this experiment. The expression of the rate is as follows:

the rate of effectiveness(%) =

$$\frac{(Group\ a) + (Group\ b)}{(the\ number\ of\ data)} \times 100.0$$

In the experiments on the system with inductive learning, the data are processed with the prediction process, the proofreading process in the case that the result is erroneous, the learning process and the feedback process, respectively. In another experiments on the system without inductive learning, the data are processed with the prediction process.

4.2 RESULTS AND CONSIDERATION

Unknown Word: chromatogram
PPW 1 "chromatogra @1, クロマトグラ @1 " (*kuromatogura*)
PPW 2 "m, ム " (*mu*)
PPW 3 "chrom @1, クロム @1 " (*kuromu*)
PPW 4 " at @1, @1 に向かって " (*ni mukatte*)
etc.
⇓
Prediction Result 1 クロマトグラ ム (*kuromato-guramu*)
etc.
Correct Target Word 1 色素列 (*shikiso-retsu*)
Correct Target Word 2 色素系 (*shikiso-kei*)
Correct Target Word 3 クロマトグラム (*kuromatoguramu*)

Figure 4: An example of prediction result in category "Learned Writing".

Table 1 and 2 show the results of this experiment. In Table 1 and 2, (A) expresses the numbers of Group a, (B) expresses the numbers of Group b, (C) expresses the rates of effectiveness, (D) expresses the numbers of Group c, (E) expresses the rates of Group c and (F) expresses the numbers of experimental data. In Table 1, the rates of effectiveness in each categories are 60.0%, 63.9%, 62.0% and 67.6%, respectively. In Table 2, rates of effectiveness in each categories are 47.8%, 45.6%, 36.1% and 47.5%, respectively.

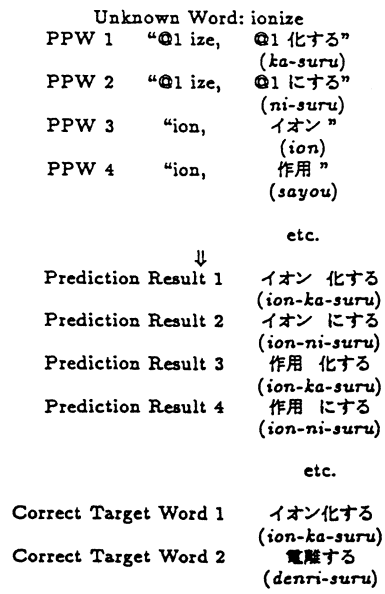


Figure 5: An example of prediction result in the category "Learned Writing".

In the category "Learned Writing," the difference of between the rates of effectiveness on two experiments is the greatest among four categories. The value of difference is 25.9 points. The reason is that the character strings of words in this category have some resemblances. Most topics of the category are research works of chemistry and some words are chemical terms. The number of chemical terms in this category is 134 and the rate of them is 35.4%. And the number of chemical terms in Group a of this category is 23 and the rate of them is 40.2%. Figure 4 shows an example of prediction result in which all PPW's are acquired from the character strings of other words. Figure 5 shows an example of prediction result in which all PPW's are given in PPW dictionary at the begining. The differences of the rates of effectiveness between two experiments are 12.2 points in the category "Press Reportage, " 18.3 points in the category "Belles, Biography, Memories" and 20.1 points in the category "Adventure and Western Fiction." These differences are smaller than the one in the category "Learned Writing." However the effective rates are almost the same among four categories in Table 1. The discussion indicates that the units are useful for our prediction process and the inductive learning is effective to acquire the units for the prediction process.

5 CONCLUSION AND FUTURE WORKS

We proposed and considered the prediction method of target words using inductive learning. In order to evaluate inductive learning, the results of experiment on the system with inductive learning were compared with the results of experiment on the system without inductive learning. The maximum difference of the rates of effectiveness is 25.9 points. We have conformed the usefulness of inductive learning on our method. In this research, we made experiments for the only derivative and compound words. However we consider our method useful for other types of unknown words. The reason is that our method uses only the information of character stings. Therefore it can adapt to other kinds of unknown words. We plan to confirm the effectiveness of our method for other kinds of unknown words in the near future.

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